

ITA0608-MACHINE LEARNING

LAB PROGRAMS

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EXP NO:3 Tree based ID3 Algorithm

AIM:

To implement and demonstrate the Tree based ID3 algorithm for finding the most specific hypothesis from the given training data

PROCEDURE:

1. Define the training dataset
2. Initialize the hypothesis with the most specific values (0).
3. Consider only the positive training examples
4. Update the hypothesis by replacing mismatched attributes with ?
5. Display the final hypothesis

PROGRAM:

1.

```
import pandas as pd
```

```
from sklearn.preprocessing import LabelEncoder
```

```
from sklearn.tree import DecisionTreeClassifier
```

```
# Dataset
```

```
data = {
```

```
    'Outlook':['Sunny','Sunny','Overcast','Rain','Rain','Rain','Overcast','Sunny'],
```

```
    'Temperature':['Hot','Hot','Hot','Mild','Cool','Cool','Cool','Mild'],
```

```
    'Humidity':['High','High','High','High','Normal','Normal','Normal','High'],
```

```
    'Wind':['Weak','Strong','Weak','Weak','Weak','Strong','Strong','Weak'],
```

```
    'Play':['No','No','Yes','Yes','Yes','No','Yes','No']
```

```
}
```

```

df = pd.DataFrame(data)

le = LabelEncoder()

for col in df.columns:

    df[col] = le.fit_transform(df[col])

X = df.iloc[:, :-1]

y = df.iloc[:, -1]

model = DecisionTreeClassifier(criterion='entropy', random_state=42)

model.fit(X, y)

sample = [[2,2,1,1]]

prediction = model.predict(sample)

print("Prediction:", "Yes" if prediction[0]==1 else "No")

```

OUTPUT:

```

Prediction: No
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but Deci
warnings.warn(

```

2

PROGRAM:

```

import pandas as pd

from sklearn.preprocessing import LabelEncoder

from sklearn.tree import DecisionTreeClassifier

data = {

    'Study':['High','High','Medium','Low','Medium','High'],

    'Attendance':['Good','Average','Good','Poor','Good','Average'],

```

```

    'Assignment':['Yes','Yes','Yes','No','No','Yes'],

    'Pass':['Yes','Yes','Yes','No','No','Yes']

}

df = pd.DataFrame(data)

le = LabelEncoder()

for col in df.columns:

    df[col] = le.fit_transform(df[col])

X = df.iloc[:, :-1]

y = df.iloc[:, -1]

model = DecisionTreeClassifier(criterion="entropy")

model.fit(X,y)

sample=[[0,2,1]]

pred=model.predict(sample)

print("Prediction:", "Pass" if pred[0]==1 else "Fail")

```

OUTPUT:

```

Prediction: Pass
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but Deci
warnings.warn(

```

3.

PROGRAM

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier

data={
'Income':['High','Medium','Low','High','Medium','Low'],
'Credit':['Good','Good','Poor','Excellent','Good','Poor'],
'Job':['Yes','Yes','No','Yes','No','No'],
'Loan':['Yes','Yes','No','Yes','No','No']
}

df=pd.DataFrame(data)

le=LabelEncoder()
for col in df.columns:
    df[col]=le.fit_transform(df[col])

X=df.iloc[:,:-1]
y=df.iloc[:, -1]

model=DecisionTreeClassifier(criterion='entropy')
model.fit(X,y)

sample=[[0,2,1]]
pred=model.predict(sample)

print("Prediction:", "Approved" if pred[0]==1 else "Rejected")
```

OUTPUT:

```
Prediction: Approved
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but Deci
warnings.warn(
```

4.

PROGRAM:

```
import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier

data={
'Experience':['High','Medium','Low','High','Medium','Low'],
'Performance':['Excellent','Good','Poor','Good','Excellent','Poor'],
'Training':['Yes','Yes','No','Yes','No','No'],
'Promotion':['Yes','Yes','No','Yes','No','No']
}
```

```

df=pd.DataFrame(data)

le=LabelEncoder()
for col in df.columns:
    df[col]=le.fit_transform(df[col])

X=df.iloc[:, :-1]
y=df.iloc[:, -1]

model=DecisionTreeClassifier(criterion='entropy')
model.fit(X,y)

sample=[[0,0,1]]
pred=model.predict(sample)

print("Prediction:", "Promoted" if pred[0]==1 else "Not Promoted")

```

OUTPUT:

```

Prediction: Promoted
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but Deci
warnings.warn(

```

5.

PROGRAM:

```

import pandas as pd
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier

data={
'Age':['Young','Young','Middle','Old','Middle','Old'],
'Income':['High','Medium','Medium','Low','High','Low'],
'Member':['Yes','Yes','No','Yes','No','No'],
'Buy':['Yes','Yes','No','No','Yes','No']
}

df=pd.DataFrame(data)

le=LabelEncoder()
for col in df.columns:
    df[col]=le.fit_transform(df[col])

X=df.iloc[:, :-1]
y=df.iloc[:, -1]

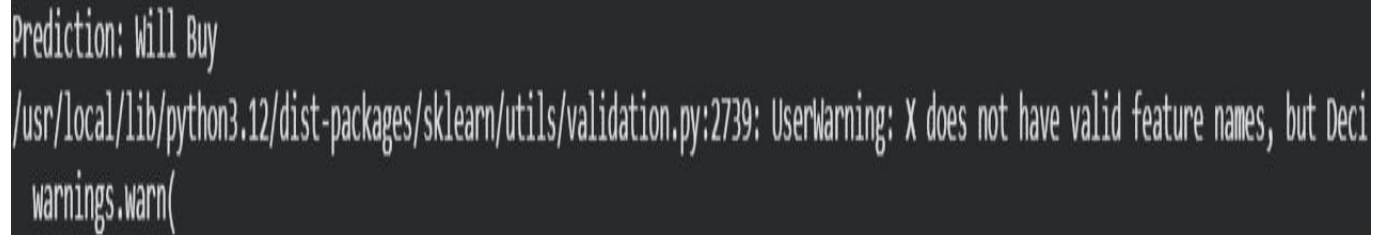
model=DecisionTreeClassifier(criterion='entropy')
model.fit(X,y)

```

```
sample=[[2,0,1]]
pred=model.predict(sample)

print("Prediction:", "Will Buy" if pred[0]==1 else "Will Not Buy")
```

OUTPUT:

A screenshot of a terminal window with a dark background. The first line shows the output 'Prediction: Will Buy'. The second line shows a warning message from sklearn: '/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but Deci'. The third line shows the start of a warning function call: 'warnings.warn('.

```
Prediction: Will Buy
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but Deci
warnings.warn(
```

RESULT :

Thus ,the tree based ID3 algorithm was implemented successfully ,and the most specific hypothesis was obtained

